

Developer Mock Test Results

Test ID:	3ec25163-0b20-42...	Student ID:	N/A
Test Type:	Developer Assessment	Date:	2026-03-04 17:27:45
Overall Score:	9/25 (36.0%)	Performance:	Needs Improvement
Warnings:	0/3	Status:	Completed

Section-wise Performance

Section	Score	Percentage	Status
Aptitude	3/10	30.0%	Needs Work
MCQ/Theory	1/10	10.0%	Needs Work
Coding	5/5	100.0%	Pass

Detailed Question Review

APTITUDE SECTION (3/10 - 30.0%)

- Q1. A product is on sale for 20% off its original price. If the original price is \$100, how much will you pay for the product after the discount?

Your Answer: \$80

Correct! Well done.
- Q2. A store buys a product for \$25 and sells it for \$30. What is the profit percentage?

Your Answer: 15%

Correct Answer: 20%

The correct answer is 20% because the profit is calculated as (selling price - cost price) / cost price * 100. The key calculation step is: (30 - 25) / 25 * 100 = 5 / 25 * 100 = 20%. The user's likely mistake was incorrect calculation, as they may have miscalculated the profit percentage, resulting in an answer of 15% instead of the correct 20%.
- Q3. A product is on sale for 25% off its original price of \$100. How much will you pay for the product?

Your Answer: \$85

Correct Answer: \$75

To find the sale price, calculate 25% of the original price (\$100) and subtract it from the original price. The key calculation step is: \$100 - (0.25 * \$100) = \$100 - \$25 = \$75. The user likely made a mistake by not correctly calculating the discount, as 25% of \$100 is \$25, not \$15, which would be the discount needed to get \$85.
- Q4. The average of 5 numbers is 12. What is the sum of the 5 numbers?

Your Answer: 55

Correct Answer: 60

■ The correct answer is 60 because the average of 5 numbers is calculated by dividing the sum of the numbers by 5. To find the sum, we multiply the average (12) by the number of items (5), so the key calculation step is: $\text{sum} = \text{average} * \text{number of items} = 12 * 5 = 60$. The user's answer of 55 is likely incorrect due to a miscalculation of this step.

■ **Q5. What is the next number: 1, 3, 6, 10, 15, ?**

Your Answer: 22

Correct Answer: 21

■ The correct answer is 21 because the pattern of the sequence is obtained by adding 2, 3, 4, 5, and so on to the previous term. The key calculation step is: $1 + 2 = 3$, $3 + 3 = 6$, $6 + 4 = 10$, $10 + 5 = 15$, $15 + 6 = 21$. The user's likely mistake was not recognizing the incremental increase in the differences between consecutive terms.

■ **Q6. A shopkeeper buys an item at Rs.100 and sells it at a 20% loss. What is the selling price?**

Your Answer: Rs.70

Correct Answer: Rs.80

■ The correct answer is Rs.80 because the shopkeeper sells the item at a 20% loss, which means the selling price is 80% of the cost price. The key calculation step is: $\text{Selling Price} = \text{Cost Price} - (20\% \text{ of Cost Price}) = 100 - (0.20 * 100) = 100 - 20 = \text{Rs.}80$. The user's likely mistake is calculating the loss amount incorrectly, as 20% of 100 is 20, not 30, which would be the case if the selling price was Rs.70.

■ **Q7. A product is on sale for 30% off its original price of \$150. How much will you pay for the product?**

Your Answer: \$100

Correct Answer: \$105

■ To find the sale price, calculate 30% of the original price of \$150 and subtract it from \$150. The key calculation step is: $\$150 * 0.30 = \45 , then $\$150 - \$45 = \$105$. The user's answer of \$100 is likely incorrect because they may have miscalculated the discount amount or subtracted it incorrectly from the original price.

■ **Q8. A shopkeeper buys 100 items at Rs.10 each and sells them at Rs.12 each. What is the profit percentage?**

Your Answer: 20%

■ Excellent! Right answer.

■ **Q9. If 10 workers can build a house in 15 days, how many workers will be needed to build the same house in 10 days?**

Your Answer: 15 workers

■ Well done!

■ **Q10. A mixture of juice and water is in the ratio 4:5. If 5 cups of juice are added to the mixture, what is the new ratio of juice to water?**

Your Answer: 5:4

Correct Answer: 9:5

■ The correct answer is 9:5 because when 5 cups of juice are added to the original mixture with a ratio of 4:5, the amount of juice increases significantly. The key calculation step is to consider the original ratio as $4x:5x$ and then add 5 cups of juice to get $(4x + 5):5x$, which simplifies to 9:5 when $x = 1$ (assuming the added 5 cups makes the juice portion equal to 9 parts). The user's likely mistake was not accounting for the increase in the juice portion correctly, leading to an incorrect ratio of 5:4.

■ **MCQ SECTION (1/10 - 10.0%)**

■ **Q1. What type of block is used to handle exceptions in Java?**

Your Answer: Catch block

Correct Answer: All of the above

■ The correct answer is "All of the above" because in Java, exception handling is typically done using a combination of try, catch, and finally blocks. The try block contains the code that might throw an exception, the catch block handles the exception if it is thrown, and the finally block contains code that is executed regardless of whether an exception was thrown.

The user's answer, "Catch block", is incorrect because while the catch block does handle exceptions, it is not the only block used in exception handling, and the question asks for the type of block "used to handle exceptions", implying a broader scope.

■ Q2. What is encapsulation in Java?

Your Answer: The concept of showing some internal details and hiding some information

Correct Answer: The concept of hiding internal details and showing only necessary information

■ The correct answer is A) The concept of hiding internal details and showing only necessary information, because encapsulation in Java is a fundamental principle of object-oriented programming that involves bundling data and methods that operate on that data within a single unit, while hiding the internal implementation details from the outside world. The user's answer, D) The concept of showing some internal details and hiding some information, is incorrect because it partially contradicts the core idea of encapsulation, which is to conceal the internal workings of an object. The user's mistake likely stems from a misunderstanding of the concept, as encapsulation is about controlling access to an object's internal state, not about partially exposing it.

■ Q3. What are the control structures in Java?

Your Answer: While loop

Correct Answer: All of the above

■ The correct answer is "All of the above" because Java has various control structures that determine the flow of a program's execution, including conditional statements like if-else, and iterative statements like for loops and while loops. The user's answer, "While loop", is incorrect because it only identifies one type of control structure in Java, whereas the language supports multiple types. The user likely mistook a specific example of a control structure for the entirety of control structures available in Java.

■ Q4. What type of software is used to write and edit Java code?

Your Answer: Compiler

Correct Answer: Code Editor or IDE

■ The correct answer is B) Code Editor or IDE because it provides a comprehensive environment for writing, editing, and managing Java code, offering features such as syntax highlighting, code completion, and project management. The user's answer, Compiler, is incorrect because a compiler is a software that translates the Java code into a machine-readable format, but it is not used for writing or editing the code. The user likely mistook the role of a compiler, which is an essential tool in the development process, but not the primary tool for writing and editing Java code.

■ Q5. What is thread safety in Java?

Your Answer: The ability of a program to execute a single thread without errors

Correct Answer: The ability of a program to execute multiple threads concurrently without errors

■ The correct answer is "The ability of a program to execute multiple threads concurrently without errors" because thread safety in Java refers to the ability of a program to handle multiple threads accessing shared resources without resulting in data inconsistencies or other errors. The user's answer is incorrect because it only considers a single thread, whereas thread safety is specifically concerned with the interactions between multiple threads. The user's likely mistake is misunderstanding the concept of thread safety, which is a crucial aspect of multithreaded programming in Java.

■ Q6. How can you compile and run a Java program?

Your Answer: Using a web browser

Correct Answer: Using a code editor or IDE

■ The correct answer, "Using a code editor or IDE", is right because Java programs require compilation and execution, which can be efficiently done using a code editor or Integrated Development Environment (IDE) like Eclipse or IntelliJ. These tools provide features such as syntax highlighting, code completion, and debugging, making it easier to write, compile, and run Java programs. The user's answer, "Using a web browser", is likely incorrect because web browsers are not designed to compile and run Java programs, they are primarily used for accessing and viewing web pages.

■ Q7. What are the types of errors in Java?

Your Answer:

Runtime exceptions

Correct Answer:

All of the above

■ The correct answer is "All of the above" because Java has three main types of errors: syntax errors, which occur when the code does not follow the language's rules; logical errors, which occur when the code does not produce the desired output due to a flaw in the program's logic; and runtime exceptions, which occur when an error happens during the execution of the code. The user's answer, "Runtime exceptions", is incorrect because it only accounts for one type of error in Java. The user likely mistook runtime exceptions as the only type of error, overlooking syntax and logical errors.

■ Q8. What type of error occurs when a program has a syntax mistake?

Your Answer: Compilation error

Correct Answer: Syntax error

■ The correct answer is "Syntax error" because it specifically refers to mistakes in the source code that violate the programming language's rules, such as missing or mismatched brackets. The user's answer, "Compilation error", is related but not exactly the same, as compilation errors can occur due to various issues, not just syntax mistakes. The user likely got it wrong because they confused the two terms, although compilation errors often result from syntax errors, they are not identical concepts.

■ Q9. What are the types of data that can be stored in a variable in Java?

Your Answer: Int, double, char, String

■ Well done!

■ Q10. How does Java handle errors?

Your Answer: Using switch statements

Correct Answer: Using try, catch, and finally blocks

■ The correct answer, "Using try, catch, and finally blocks", is right because Java utilizes these blocks to handle errors and exceptions in a structured and organized manner. The try block contains the code that may potentially throw an exception, the catch block handles the exception if it occurs, and the finally block executes regardless of whether an exception was thrown. The user's answer, "Using switch statements", is likely incorrect because switch statements are used for conditional branching based on specific values, not for error handling.

■ CODING SECTION (5/5 - 100.0%)

■ Q1. Write a Java program that reads an array of integers from stdin and sorts it in ascending order.

****Input Format:** - Line 1: The number of elements i...**

Your Answer:

```
import java.util.Scanner;
import java.util.Arrays;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        sc.close();

        Arrays.sort(arr);

        for (int i = 0; i < n; i++) {
            System.out.print(arr[i]);
            if (i < n - 1) {
                System.out.print(" ");
            }
        }
    }
}
```

```

    }
  }
}

```

■ Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and effective implementation. Great job on achieving an "Accepted" status, indicating that your solution meets all the requirements.

■ **Q2. Write a Java program that reads a string from stdin and counts the frequency of each word. ****Input Format:**** - Line 1: A string ****Output Format:**** -...**

Your Answer:

```

import java.util.Scanner;
import java.util.LinkedHashMap;
import java.util.Map;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String input = sc.nextLine();
        sc.close();

        String[] words = input.trim().split("\\s+");

        Map<String, Integer> frequencyMap = new LinkedHashMap<>();

        for (String word : words) {
            frequencyMap.put(word, frequencyMap.getOrDefault(word, 0) + 1);
        }

        for (Map.Entry<String, Integer> entry : frequencyMap.entrySet()) {
            System.out.print(entry.getKey() + ":" + entry.getValue() + " ");
        }
    }
}

```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as intended. Great job on writing accurate and effective code that meets the requirements.

■ **Q3. Write a Java program that reads a string from stdin and validates it using a regular expression. ****Input Format:**** - Line 1: A string ****Output Format:**** -...**

Your Answer:

```

import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String input = sc.nextLine();
        sc.close();

        // Regex: only lowercase letters
        String regex = "^[a-z]+$";

        if (input.matches(regex)) {
            System.out.println("Valid");
        } else {
            System.out.println("Invalid");
        }
    }
}

```

```
}
```

■ Your code has successfully passed all test cases, indicating a correct and effective implementation of the solution. Great job on achieving an "Accepted" status, keep up the good work!

■ **Q4. Write a Java program that reads an array of integers from stdin and finds the maximum element.**

****Input Format:** - Line 1: The number of elements in ...**

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        sc.close();

        int max = arr[0];

        for (int i = 1; i < n; i++) {
            if (arr[i] > max) {
                max = arr[i];
            }
        }

        System.out.println(max);
    }
}
```

■ Your code has successfully passed all test cases, indicating a correct and effective implementation of the solution. Great job on achieving an "Accepted" status, keep up the good work!

■ **Q5. Write a program that reads a string and prints the string with all characters converted to uppercase, except for the first character which should be i...**

Your Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String input = sc.nextLine();
        sc.close();

        if (input.length() == 1) {
            System.out.println(input);
        } else {
            System.out.println(input.toLowerCase() + " WORLD");
        }
    }
}
```

■ Your code has successfully passed all test cases, demonstrating a thorough understanding of the problem and its solution. Great job on achieving an "Accepted" status, indicating that your code is correct and efficient.

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.